

Ayush Raj

Third Year Undergraduate
Department of Aerospace Engineering

✉ ayushraj23@iitk.ac.in | 📞 +91-6901768815
🌐 Ayush Raj | 📍 Ayushraj6942

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	6.6/10
2023	CBSE(XII)	Indian Convent School,Kota	94.2%
2021	CBSE(X)	Army Public School,Sri Ganganagar	96.4%

Scholastic Achievements

- Secured All India Rank **3946** in **JEE Advanced 2023** among 2.5 lakh shortlisted candidates.
- Secured All India Rank **4175** in **JEE Mains 2023** among 11 lakh candidates.

Research Experience

SURGE’25 IIT KANPUR | B2B Brand Equity | Prof.RN Sengupta (May’25-July’25)

Objective	Identify determinants of B2B brand equity using literature, company data, and test models via ML .
Approach	- Reviewed literature on B2B Brand Equity to map constructs like awareness, loyalty, trust, and reputation. - Applied SEM and multiple logistic regression on 5 firms; analyzed impact of digital marketing using MROI. - Analysed Semantic Brand Score to measure brand importance and strength using textual data and Monte Carlo Simulation for Brand Equity Risk Optimization
Impact	Proposed methods to quantify brand equity using ML models and highlighted digital marketing’s role via MROI .

Key Projects

Hydro Mechanics Of Finvision | Google Developer Group Project 🔄 (Dec’24-Mar’25)

- Built real-time financial data pipelines via **Selenium, Requests, BeautifulSoup** from **Yahoo Finance, NSE**, etc.
- Applied **Black-Scholes, Binomial, Vasicek, CIR, Merton Models**; simulated **GBM, ADF, Weiner** processes for forecasting & risk.
- Used **ML (LSTM, RNN)** for **ETH** prediction; explored quantum models using **TensorFlow Quantum, PennyLane**.

White Paper For Assistive Devices | IITK Consulting Group | Department Of SnT 🔄 (April’25-June’25)

- Worked under **Dr. Anita Aggarwal (DST, NAMEDD)** on a national white paper for assistive devices (ADs), targeting policy & investment.
- Co-authored recommendations using global best practices comparative insights from **USA, Japan, Germany, Denmark**.
- Built global leaderboard, mapped **India’s top 10 medtech states**; analyzed **7+ govt schemes, 20+ VC firms/accelerators/startup studios**.

Nuvé: AI-Driven Fashion Trend Forecasting | Course Project-DES646 | Prof. A.K.Behera 🔄 (Sep’25-Nov’25)

- Developed an **AI-driven visual analytics system** for fashion trend forecasting as part of **DES646: AI/ML for Designers** at IIT Kanpur.
- Performed **sentiment analysis using NLP** on user comments and combined engagement metrics (likes, comments) to compute a composite **trend score**.
- Designed an interactive **Streamlit + Plotly dashboard** enabling designers to explore clusters, sentiment distributions, and emerging fashion patterns in real time.

Technical Skills

- Programming Languages:** C, C++, Python
- Software:** Canva, MS Excel, Jupyter
- Libraries:** Numpy, Pandas, Tensorflow, Seaborn, Matplotlib, Keras, Scikit-Learn

Relevant Courses

Computer-Aided Decision System	Fundamentals of Computing	AI/ML for Designers
--------------------------------	---------------------------	---------------------

Positions of Responsibility

Business Head | IITK Rocketry & Space Exploration Team Mar’24-Present

- Contributed to **avionics, structures, propulsion, recovery**; launched rocket to **600 ft** apogee in **July 2024**.
- Managed finances; secured **2 sponsorships**, handled **5 lakh** budget, and oversaw bill clearance.
- Led **30+ juniors**, handled outreach, sponsorship, and participating as Recovery member in **INSPACE Competition**.

Marketing Secretary | Udghosh’24 Aug’24-Oct’24

- Led lead generation and **database management** for **100+ companies** to support sponsorship and marketing outreach.
- Developed strong **communication skills** including cold calling, email writing, and persuasive brand negotiation.

Extra-Curricular Activities

- Received **2 LORs and certificates** for outstanding performance in **finlatics financial markets & investment banking** project.
- Received **certificate** for participating in **IIT Kanpur certification program on Python For AI,ML and DL**.
- Completed **NCC ‘C’ Certificate** program, demonstrating leadership, discipline, and endurance.
- Cleared **PDR and CDR** rounds of **INSPACE Model Rocketry Competition** with IITK RaSET team.